Building Systems for Scalable and Responsible Data Management

The Data Intelligence Loop

Zhengyi Yang
University of Sydney

zhengyi.yang@sydney.edu.au
Office: Room 350, Building J12

Motivation

> My research asks how to build data systems that scale to complex workloads, support reliable AI workflows over data, and inform responsible real-world decision-making.

Complex and dynamic data	Tables, graphs, text, documents, multimedia data, logs, streams, and evolving workloads.
AI data workloads	Retrieval, indexing, evidence selection, reasoning, verification, serving, and cost control.
Real-world applications	ESG, finance, workforce analytics, fairness, climate, health, and scientific discovery.

Database systems provide the foundation for managing, processing, and reasoning over data across these motivations.

02 / 21

Three-Theme Research Agenda

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STAGE

Data Processing
Process heterogeneous and dynamic data efficiently.

↓

Data Understanding
Extract structure, semantics, context, and evidence.

↓

Data Intelligence
Support responsible decisions in real domains.

RESEARCH FOCUS

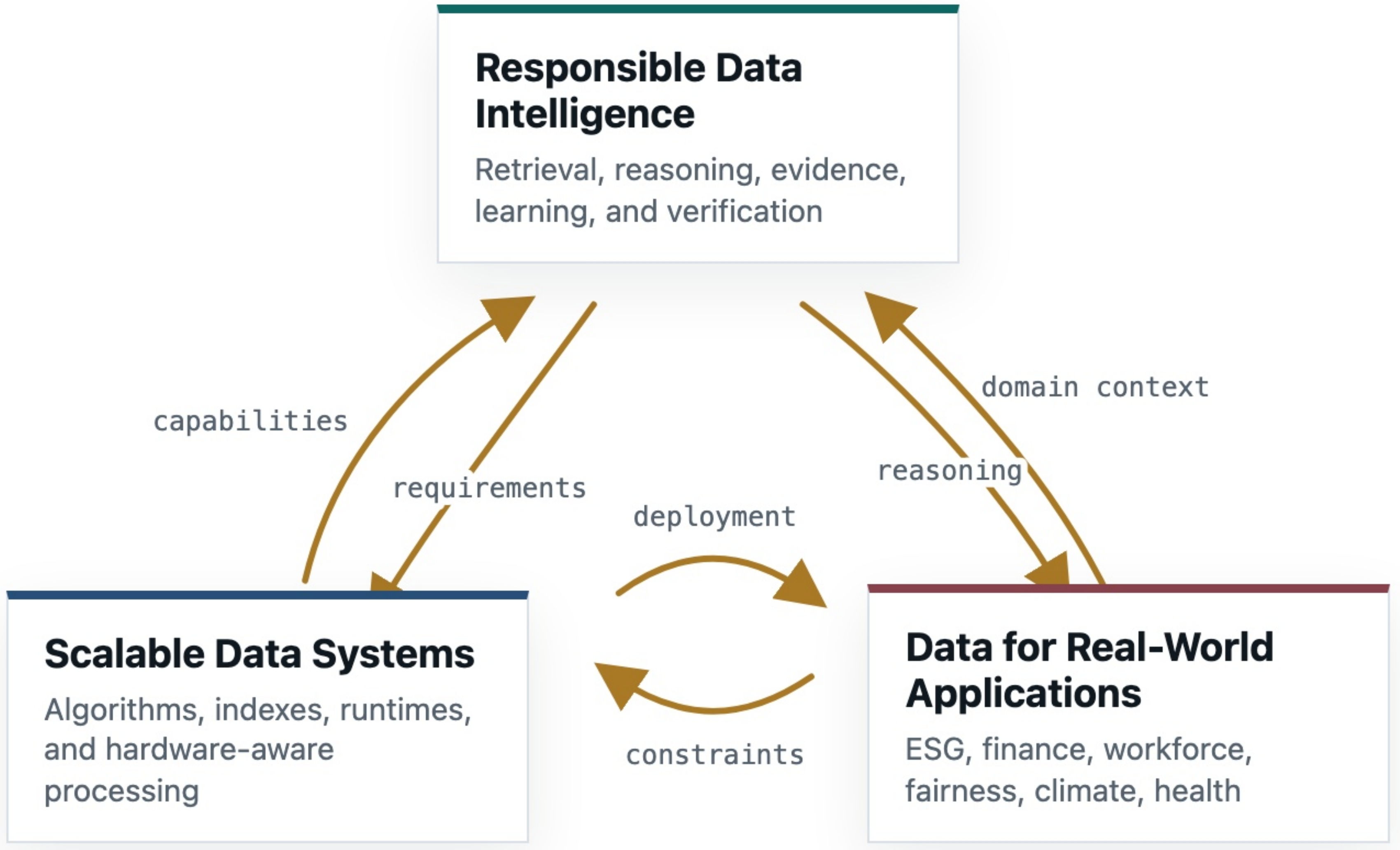
THEME 1
Scalable Data Systems
Graphs, tables, streams, query processing, GPU, out-of-core

THEME 2
Responsible Data Intelligence
GraphRAG, LLM reasoning, AI/ML for DB, tabular learning, data-centric AI

THEME 3
Data for Real-World Applications
ESG, finance, built environment, fairness, climate, health

The agenda links scalable systems, responsible intelligence, and real-world applications through three themes.

The Data Intelligence Loop



BOTTOM-UP

Systems enable intelligence

Efficient data systems organize and process complex data, so AI methods can retrieve evidence, reason over it, and support decisions.

TOP-DOWN

Applications define requirements

Real-world use cases tell us what the systems must guarantee: traceable evidence, reliable outputs, fair decisions, validation, and deployment readiness.

Processing enables understanding; understanding enables intelligence; real-world applications feed requirements back into data systems.

PART I

Scalable Data Systems

Data processing is about making complex and dynamic data computationally manageable at scale.

Graph Analytics and Processing

Graphs, hypergraphs, temporal graphs, matching, centrality, decomposition.

Database Systems & Query Processing

Query processing, joins, JIT/runtime, OLTP kernels, storage, transactions.

Parallel and Hardware-Aware Processing

GPU, FPGA, RISC-V, UMA, streaming, parallelism, memory- and I/O-aware execution.

Graph Analytics at Scale

Complex relational structures are no longer only graphs; they include higher-order and temporal interactions that make naive enumeration infeasible.

REPRESENTATIVE TOPICS

- **Higher-order graph and hypergraph** analytics
- **Pattern matching** and **substructure search**
- **Connectivity and path** analytics
- **Temporal and evolving graph** analytics

Turn structural properties into indexes, pruning rules, and execution strategies that avoid exhaustive search.

SELECTED WORK

COMPLETED

Nucleus decomposition
SIGMOD 2026

Diversified graph pattern matching
VLDB 2026

Hypergraph matching
ICDE 2026 · ICDE 2023

Hypergraph decomposition
VLDBJ 2026 · SIGMOD 2025 · VLDBJ 2025 · CIKM 2024

Constrained path analytics
ICDE 2025

Clique and dense subgraph analytics
WWW 2025

Temporal graph analytics
WWW 2024

Graph pattern matching systems
SIGMOD 2021 · ICDE 2021 · VLDB 2019

UNDER SUBMISSION

Higher-order graph analytics
SPRINGER MONOGRAPH, IN PRESS

Nucleus and nuclear decomposition
SIGMOD 2027 · VLDB 2027

Hypergraph core maintenance
ICDE 2027

Reachability indexing and querying
EDBT 2026 UNDER REVISION

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Database Systems & Query Processing

Database performance depends not only on algorithms but also on how query logic interacts with compilation, runtime execution, storage, and deployment constraints.

REPRESENTATIVE TOPICS

- **Transactional engine** design
- **Query processing**, joins, and **runtime optimization**
- **JIT / MLIR** compilation for **database execution**
- **Storage**, **concurrency**, and **recovery**

Systems optimization should make database engines faster while keeping them maintainable and deployable.

EXECUTION PIPELINE

01 **Query** Declarative intent and workload shape. → 02 **Runtime** JIT, MLIR, and specialized execution. → 03 **Engine** Storage, transactions, and deployment.

SELECTED WORK

SPOTLIGHT

PhoebeDB: High-Performance Disk-Based RDBMS Kernel for OLTP

EDBT 2025 INDUSTRY | INDUSTRY COLLABORATION | TARGET: INDUSTRY-READY

- **Industry collaboration** toward an **enterprise-oriented**, industry-ready RDBMS kernel.
- **PostgreSQL-compatible OLTP** on **commodity hardware**, reducing migration and deployment risk.
- **Full-stack kernel design**: temperature-aware storage, coroutine runtime, smart scheduling, transaction management, WAL, and snapshots.
- Reports **13.7M tpmC** and **30M tpm** on **TPC-C** using one machine, with **27x** improvement over PostgreSQL.

COMPLETED

Batch-dynamic high-dim kNN join
ADMA 2024 BEST STUDENT PAPER

UNDER SUBMISSION

Fully dynamic high-dim kNN joins
VLDBJ

Parallel and Hardware-Aware Processing

At billion scale, performance depends on coordinating distributed execution, parallelism, memory hierarchy, storage, and accelerators.

REPRESENTATIVE TOPICS

- **Distributed** and out-of-core processing
- **Parallel runtime** and **adaptive scheduling**
- **GPU-/FPGA-accelerated indexing** and analytics
- **Emerging hardware** and **scalable execution**

Scale by co-designing algorithms, runtimes, and hardware-aware execution.

EXECUTION DIMENSIONS

Distributed execution | Memory / Out-of-core | Parallel runtimes | GPU / FPGA / RISC-V

SELECTED WORK

COMPLETED

Out-of-core graph processing system
SIGMOD 2026

GPU-accelerated vector index construction
SIGMOD 2026

GPU-accelerated range top-k query
HPDC 2026

Database benchmarking on RISC-V
VLDB ADMS 2025

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FPGA-accelerated graph random walks
SIGMOD 2027

SIMD set-intersection benchmarking
SIGMOD 2027

Out-of-core vector search and indexing
EUROSYS 2027

Unified-memory DataFrame processing
VLDB 2026 DEMO

PART II

Responsible Data Intelligence

Data understanding is about turning processed data into structure, semantics, context, and reliable evidence for AI systems.

Data Systems for Reliable LLMs

Retrieval, evidence selection, reasoning, agent memory, and context management.

AI/ML for Databases and Data Systems

Cost estimation, tuning, temporal analytics, and costly system decisions.

Data-Centric AI

Relational, unstructured, and graph data; efficient learning.

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Data Systems for Reliable LLMs

Reliable LLM applications over data need systems for retrieval, evidence organization, verification, reasoning orchestration, and serving, not just fluent generation.

REPRESENTATIVE TOPICS

- **Data-to-structure** construction
- **Evidence retrieval** and **selection**
- **Reasoning** and **verification**
- **Serving**, **memory**, and **context management**
- **LLM safety** and **jailbreaking**

Reliable LLMs create data-management workloads: indexing, planning, provenance, cost control, and evidence quality.

RETRIEVAL-TO-REASONING PIPELINE

01 **Corpus** Raw text, tables, documents. → 02 **Graph** Structure and semantic units. → 03 **Evidence** Retrieval and verification. → 04 **Reasoning** Serving and memory.

SELECTED WORK

COMPLETED

Adaptive GraphRAG retrieval
KDEXLLM 2026 BEST PAPER

LLM-graph for data exploration
SIGMOD DEMO 2025

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Graph CoT reasoning and serving
TOIS

Cost-efficient KG construction
TKDE

Semantic-unit retrieval for HyperRAG
EMNLP 2026

Long-term memory for LLM agents
EMNLP 2026

Memory-driven LLM jailbreaking
EMNLP 2026

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AI/ML for Databases and Data Systems

This line of work learns to approximate, tune, and adapt costly decisions in data systems while respecting query, graph, and temporal semantics.

REPRESENTATIVE TOPICS

- **Learning** for **query optimization**
- **Graph neural networks (GNNs)**
- **Representation learning** over **structured data**
- **Deployment-aware** learned components

AI/ML is useful when learning acts as a system component with clear semantics, error behavior, and operational cost.

Query Optimization
Cost estimation and optimization signals for expensive database decisions.

Graph Analytics
Temporal centrality and graph signals where exact computation is costly.

SELECTED WORK

COMPLETED

Learning-based temporal centrality
WWW 2026 WWW 2024

Adaptive parallelism tuning for stream processing
ICDE 2025

Heterogeneous graph learning
CIKM 2025

UNDER SUBMISSION

Learned query cost estimation
ICDE 2026

Event-centric HIN learning
ICDM 2026

Learning-based attributed community detection
TKDD

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Data-Centric AI

Data-centric AI studies how relational, unstructured, and graph data shape model learning, reasoning, efficiency, and reliability.

REPRESENTATIVE TOPICS

- **Structured data representation** and **reasoning**
- **Data condensation** and **efficiency**
- **Label quality** and **aggregation**
- **Data quality** for **AI workflows**

Data-centric AI brings classic data quality questions back into the core of model performance and reasoning reliability.

DATA PROBLEMS

Relational data
Tables, schemas, joins, and structured reasoning.

Unstructured data
Text, documents, and multimodal evidence.

Graph data
Entities, relations, paths, and context.

Efficiency
Less data, lower cost, and faster learning.

SELECTED WORK

COMPLETED

Tabular data condensation
ICDE 2026

Structured table reasoning
WWWJ 2025

UNDER SUBMISSION

LLM-based tabular reasoning
EMNLP 2026

Label aggregation in crowdsourcing
ICDE 2026

PART III

Data for Real-World Applications

Data intelligence is about using processed and interpreted data to support responsible decisions in real-world domains.

ESG and Finance

Disclosure, standards, fraud, interpretable analytics.

Built Environment

Smart construction, careers, leadership, guidance.

Fairness and Diversity

Representation, bias, evaluation, context.

Science and Society

Climate, health, materials, validation.

ESG and Financial Intelligence

ESG (Environmental, Social, and Governance) and finance turn messy documents, reports, and behavioral traces into decisions that require traceability and interpretation.

REPRESENTATIVE TOPICS

- **Document and report** intelligence
- **Knowledge-grounded evidence** management
- **Financial behavior and risk** analytics
- **Interpretable** decision support

The database problem is evidence governance: linking claims, standards, provenance, and models into auditable workflows.

ESG

Standards, reports, provenance, auditability.

Finance

Noisy behavior, robust signals, fraud detection.

SELECTED WORK

COMPLETED

Agent-based ESG report analysis
ICSA 2026

Finance analysis and fraud detection
BIGCOMP 2026 ADMA 2025

UNDER SUBMISSION

ESG data management survey
THE INNOVATION

Standard-aware ESG disclosure analysis
ICDM DEMO 2026

Mining ESG analysis
APWEB-WAIM

ESG knowledge graph construction
ICKG 2026

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ESG Data Management

AI-Empowered ESG Data Analytics in Responsible Investment

The project leverages advanced AI and data management to improve ESG risk assessment, performance forecasting, and report generation.

BACKGROUND

ESG links environmental impact, social responsibility, and governance quality to investment, compliance, and accountability.

SIGNIFICANCE

Reliable data infrastructure makes ESG claims traceable, comparable, and actionable for responsible investment.

FOR 4605 - Data Management and Data Science

ESG analysis is a data-management problem before it is an AI problem: fragmented standards, disclosures, market signals, enterprise evidence, risk signals, forecasts, and reports must become timely, objective, and auditable decision support.

PROJECT AIMS	Assess	Forecast	Generate
	Improve ESG risk analysis from complex, mixed-source data.	Surface performance signals for responsible investment.	Produce transparent reports backed by auditable evidence.

NEED -> OUTPUT -> DATA -> SYSTEM

National need
Reliable ESG insights support sustainable investment, corporate responsibility, and regulatory compliance.

Expected outcome
More accurate, accessible, timely, and objective ESG analysis for investors and companies.

Data science challenges
Availability, quality, integration, traceability, timeliness, privacy, and cross-standard semantics.

Systems agenda
ESG knowledge layer, claim-level provenance, retrieval/reasoning guardrails, and workflow evaluation.

GOVERNED ESG DECISION LOOP

Report
Transform standards, disclosures, and evidence into structured ESG claims.

→

Review
Check completeness, consistency, standard alignment, and evidence quality.

→

Respond
Support risk, compliance, investment, and operational decisions with audit trails.

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Built Environment Intelligence

Built-environment and smart-construction applications transform fragmented professional, project, and regulatory traces into interpretable evidence for career, compliance, sustainability, and organizational decisions.

REPRESENTATIVE TOPICS

- **Built-environment** decision support
- **Workforce and career** intelligence
- **Compliance and sustainability** analytics
- **Heterogeneous domain data** integration

Real-world decision support often begins as entity resolution, evidence extraction, and schema design over noisy human-centered data.

DATA TO DECISION FLOW

Professional traces → Structured evidence → Decision support

SELECTED WORK

COMPLETED

Construction career guidance
BIGCOMP 2026 · ISARC 2026

Built-environment GraphRAG systems
BUILDINGS 2026

Workforce leadership and progression analysis
BUILDINGS 2026 · CRIOCM 2026 · IJCM 2025

UNDER SUBMISSION

Australian built-environment AI co-pilot
AUTOMATION IN CONSTRUCTION

Compliance and sustainability intelligence
ARC LINKAGE PROJECT

Fairness and Diversity Intelligence

Fairness work requires careful data modeling because representation, missingness, and measurement choices shape downstream conclusions.

REPRESENTATIVE TOPICS

- **Representation** and **measurement**
- **Fairness** in decision support
- **Diversity and progression** analytics
- **Evaluation** of **model behavior**

Fairness is not only a model metric; it is also a data-management problem of measurement, provenance, and context.

FAIRNESS QUESTIONS

Who is underrepresented?
Groups missing or poorly measured.

Who is affected?
Real decisions shaped by outputs.

What harm can scale?
Bias amplified across workflows.

What is the social impact?
Fairness, opportunity, and trust.

SELECTED WORK

● COMPLETED

Gender representation and career progression
JCEM 2026 · IJCM 2025 · IJCM 2025

LLM fairness and bias evaluation
FAIML 2026 · AI 2025

● UNDER SUBMISSION

LLM bias in hiring
WORK IN PROGRESS

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Scientific and Societal Intelligence

Scientific and societal applications need AI pipelines that respect domain constraints and produce outputs that can be validated.

REPRESENTATIVE TOPICS

- **Domain-aware** scientific modeling
- **Health and conversation** intelligence
- **Scientific discovery** workflows
- **Validation** and **trustworthy outputs**

Domain-facing intelligence needs data systems that make heterogeneous evidence scalable, checkable, and reusable.

DOMAIN PILLARS

Climate

Heterogeneous evidence and domain-aligned modeling.

Health

Conversation data, semantics, and responsible interpretation.

Science

Materials discovery and validation-centered AI pipelines.

SELECTED WORK

UNDER SUBMISSION

Weather and climate modeling

NEURIPS 2026

Health conversation data intelligence

ADMA 2026

AI-assisted materials discovery

MULTIPLE MANUSCRIPTS

SYNTHESIS

What This Agenda Enables

A database systems agenda for AI-era decision-making.

The common thread is to turn complex data into scalable computation, reliable understanding, and responsible decision support in real domains.

SYSTEMS | AI PIPELINES | APPLICATIONS

CAPABILITY

Scalable data substrates

Algorithms, indexes, runtimes, and hardware-aware execution for complex workloads.

RELIABILITY

AI-facing data understanding

Retrieval, reasoning, provenance, and verification over structured and unstructured data.

IMPACT

Domain-grounded intelligence

Real-world settings feed back requirements for fairness, robustness, auditability, and deployment.

The broader contribution is to connect database systems, AI pipelines, and real-world data deployment through one coherent data intelligence loop.

Research Roadmap

DIRECTION 1

Data Systems for Reliable LLMs

- RAG-aware indexing and retrieval planning
- Evidence-aware query processing and verification
- Cost-based optimization for reasoning pipelines
- Provenance and audit trails for LLM outputs

DIRECTION 2

Scalable Processing for Complex Data

- Distributed graph, hypergraph, and temporal analytics
- Hardware-aware vector and graph indexing
- Adaptive parallel runtimes for evolving workloads
- Emerging architectures for scalable execution

DIRECTION 3

Responsible Data Intelligence

- ESG and financial data intelligence
- Built environment and smart construction
- Fairness, diversity, and accountable data use
- Climate, health, and scientific data intelligence

INDEXING | OPTIMIZATION | RUNTIME SYSTEMS | DISTRIBUTED PROCESSING | DATA QUALITY | PROVENANCE | BENCHMARKS

Q&A

Collaboration & Discussion

RESEARCH SUMMARY

Scalable Data Systems
Graph and hypergraph analytics, database runtime, streaming, and hardware-aware processing.

Responsible Data Intelligence
GraphRAG, LLM retrieval and reasoning, AI4DB, tabular learning, and data-centric AI.

Data for Real-World Applications
ESG, finance, built environment, fairness, climate, and health.

POSSIBLE COLLABORATIONS

RDBMS and HTAP systems
OLTP/HTAP engines, query execution, isolation and consistency, runtime optimization, and provenance.

Data engineering for AI and science
ML-in-DB, multicore/cloud execution, scientific data management, caching, and benchmarking.

Graph data management at scale
Higher-order and temporal graphs, subgraph matching, dynamic graph processing, and GraphRAG indexing.

Thank you!

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